

GRUB

The GRand Unified Boot loader is probably the slickest Linux boot loader of all time. Historically, it's part of the GNU project, and is more accurately called GNU GRUB. It comes with a neat menu interface that features an easy-to-use menu editor, as well as a command-line interface. GRUB is also a shade easier to use than LILO, and is recommended for new Linux users, especially if you plan to multi-boot with Windows XP. Red Hat has made GRUB the default boot loader, and there are indications that Red Hat plans to drop LILO in the near future. Installation is simple, and GRUB can load any operating system easily.

You can install GRUB directly on the MBR, and it will boot both Windows 98 and Windows XP. Unlike LILO, you won't need to use NTLDR to boot into GRUB, but you can get NTLDR to load GRUB if you want. Setting this up is a procedure similar to that described for LILO. This way, the MBR remains safe and GRUB resides in the first sector of the boot partition.

Here's a quick look at what GRUB gives you: GRUB works off the **'/etc/grub.conf'** configuration file. Unlike LILO, GRUB will read the configuration file at boot time, so you don't need to re-run GRUB every time you make a change. You can edit the menu selections to enter



Editing your GRUB menu is as easy as pressing [E] at the prompt

parameters at boot time. Select the menu entry and press [E]—this opens the menu editor. Make the required changes, and

press [Enter] to register them. For example, to boot into single user mode, select the kernel you want to boot into, then press [E]. Select the line that starts with the command 'kernel' and add the word 'single' to the end. Now you should be able to boot into Linux in single user mode. The line should look like **'kernel /vmlinuz-2.4.18-14 ro root=LABEL=/ single'**.

For further configuration, check out the command line feature with the GRUB prompt. To access this, press [C]. [Tab] will give you a list of commands here. You can also boot the kernel from here. For example, the following command will boot into the kernel image called 'newkernel' with the root drive being **'/dev/hda5'**:

```
'kernel (hd0,1)/boot/newkernel
root=/dev/hda3 read only'
'boot'
```

Press [Enter], and if the kernel image is located in the right place, the kernel will boot. GRUB is simple, elegant and undoubtedly a very good boot loader.

'default=linux' indicates that the default OS to boot into is Linux. Remember, don't edit the 'boot', 'map', 'install' and 'message' values—all of these are set by the LILO installer, and do not need to be changed.

'image=/boot/vmlinuz-2.4.18-14' is the line that tells LILO what kernel to load. This file is located in the **/boot** directory and is the core kernel image file. The lines that follow are the additional parameters that are passed on to the kernel while booting. The 'label' is the name of the kernel; this is the name that will be displayed in the boot menu, and is used next to the 'default' parameter. 'read-only' means that the root filesystem is initially mounted as read-only, and later in the boot process, is made read-write. If you compile a new kernel, you will need to make similar entries in the lilo.conf file, with parameters specific to the newly compiled kernel.

The 'other' entry, and the lines that follow, let LILO boot to a non-Linux operating system. In this case, no kernel image file is loaded; instead, the boot sector of the specified partition is loaded into memory and executed.

Booting Windows

Booting into Windows 98 is simple, as LILO will load and execute the specified Windows drive boot sector, as indicated in lilo.conf. However, LILO cannot directly

boot into Windows XP, as Windows XP uses the NT boot loader, NTLDR. To be able to successfully multi-boot Windows XP and Linux with LILO, you will need to tweak NTLDR. This way, you can boot into Windows XP or Linux easily. To boot into Linux, you have to get NTLDR to transfer control to LILO. Remember, this technique works only if you run Windows XP on the FAT32 filesystem; LILO does not work with NTFS.

In our example, Windows XP resides on the **/dev/hda1** partition, the **/boot** partition is **/dev/hda2** and the **/(root)** partition is **/dev/hda5**. First, you need to create an image of the Linux boot partition. As root, enter the following command:

```
dd if=/dev/hda2 of=/bootsect.lnx bs=512
count=1
```

This command copies the first 512 bytes of the **/boot** partition into a file—**bootsect.lnx**, in the **/(root)** partition.

Next, you need to copy this file to the Windows XP drive. You can do this either by copying the bootsect.lnx file into a floppy, booting into Windows XP and copying it to the 'C:' drive, or by mounting the XP filesystem using the 'mount' command:

```
mkdir /windowsxp
```

This creates a directory, a mount point for

Windows XP. Now mount the drive using the command:

```
mount -t vfat /dev/hda1 /windowsxp
```

Copy bootsect.lnx to the C drive by typing:

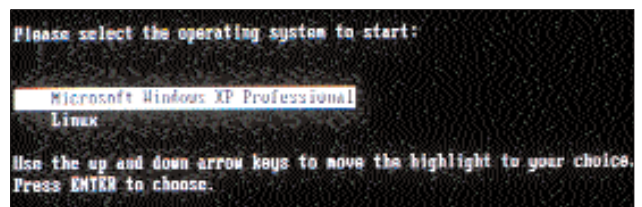
```
cp /bootsect.lnx /windowsxp
```

Now you need to boot into Windows XP to tweak the NTLDR. You can do this by adding an entry into the **boot.ini** file—the NTLDR equivalent of the **lilo.conf** file. Normally, this is not visible in the C drive. You can edit it by going to **Start > Run** and typing in **cmd.exe**, and at the command prompt typing **edit boot.ini**. Add the following line to the end of the file:

```
C:\bootsect.lnx="Linux"
```

Now, save the file and close it. Do not edit any other settings, or you may not be able to boot into Windows XP. The edited file should look like this:

```
[boot loader]
timeout=30
default=multi(0)disk(0)rdisk(0)partition(1)\WINDOWS
```



NTLDR boot menu: Booting into Linux, the Windows XP way